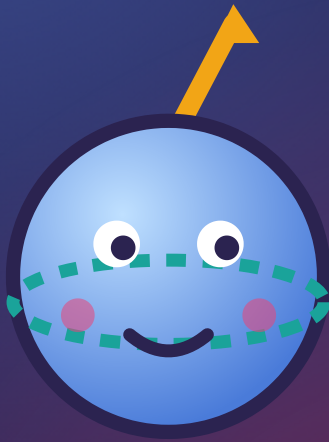


A NOTEBOOK FIELD GUIDE

Four Is Too Many

...two of them are fake.



Spotting, naming and gently befriending
the wild qubit

from the notebook of Sharon Epic Armando

Charlottetown, PEI · quantumspatial.lol · autumn 2026

BEFORE WE SET OFF

Hi there, explorer

This little guide began life as a spiral notebook: pages of handwriting, arrows, crossings-out and one rather good drawing of a chip. Everything here is from those pages, tidied up, coloured in and occasionally decorated with stickers.

You will need: curiosity, a cup of tea, and a willingness to throw away numbers that turn out to be fake.



Hello! I'm Blochy.

I'm a qubit, and I live on a sphere (you'll find out why). I'll pop up in the margins with little notes, like a friend leaning over your shoulder. Nothing scary, I promise.



Whenever a number turns out not to matter, it gets stamped.

What's inside

- 1 Zero, One, and the Bit in Between
- 2 Four Is Too Many
- 3 Welcome to the Globe
- 4 Why $\theta/2$? The Great Stretch
- 5 Two's Company: Entanglement
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- 8 The Hardware Zoo
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- ✦ Epilogue: Hi There, Qubit

Zero, One, and the Bit in Between

A classical bit is either 0 or 1.

That's it. A light switch, a coin lying flat on the table, a yes or a no. Every spreadsheet, every web server and most of what you do every day is built out of these two little answers.

A qubit can be written as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

where α and β are **complex amplitudes**. Until you measure it, the qubit is in a superposition of both. Not "secretly 0 or secretly 1, we just don't know", but a genuine blend with a size and a direction for each part.



Bloch's margin note

Tiny detail: I'm only in a superposition when *both* α and β are non-zero. If $\beta = 0$, I'm just plain $|0\rangle$, sitting at the North Pole being perfectly ordinary. Which is allowed. Everyone needs a quiet day.

The amplitudes are not probabilities...

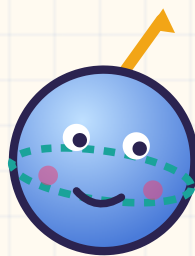
...but they make them. Square their sizes and you get the odds of each outcome when you look:

$$P(0) = |\alpha|^2, \quad P(1) = |\beta|^2$$

The Physics of One Qubit

The plan:

- ✦ *state*
- ✦ *Bloch sphere*
- ✦ *measurement*
- ✦ *phase*



A single qubit is just a two-level quantum system you can control.

Two energy levels; you pick them and call them $|0\rangle$ and $|1\rangle$. It might be two levels of an atom, two states of a tiny superconducting circuit, or two polarisations of light. The universe offers plenty of candidates. What makes one a *qubit* is that you can steer it.

Everything else is how that system sits between those two levels.

The full state is

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \text{with the only hard rule} \quad |\alpha|^2 + |\beta|^2 = 1.$$

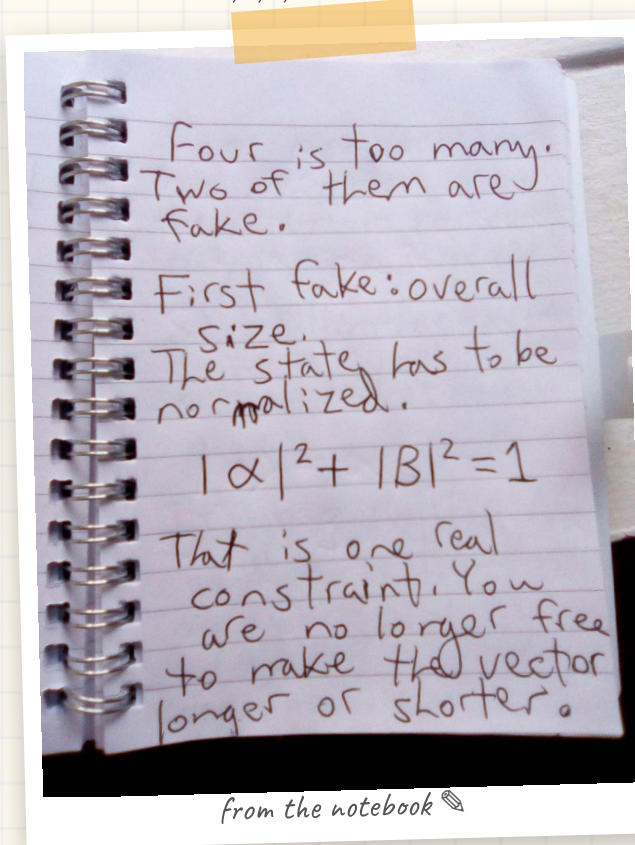
Those two complex numbers look like four real numbers. And that is where our story really begins.

Four Is Too Many

Start with the raw object. Each complex amplitude has a real part and an imaginary part:

$$\alpha = a + ib, \quad \beta = c + id$$

So that's four real numbers: a, b, c, d . Four dials to twiddle. Except...



"Four is too many.
Two of the are fake."

Unmasking the Fakes

First fake: overall size

The state has to be normalised:



$$|\alpha|^2 + |\beta|^2 = a^2 + b^2 + c^2 + d^2 = 1$$

That is one real constraint. You're no longer free to make the vector longer or shorter. Four numbers minus one constraint leaves three. Geometrically, that is the **3-sphere**, S^3 , sitting inside $\mathbb{C}^2$ (which is really $\mathbb{R}^4$ in a trench coat).

Second fake: overall phase

If you multiply the whole state by a phase,



$$|\psi\rangle \longrightarrow e^{i\gamma}|\psi\rangle$$

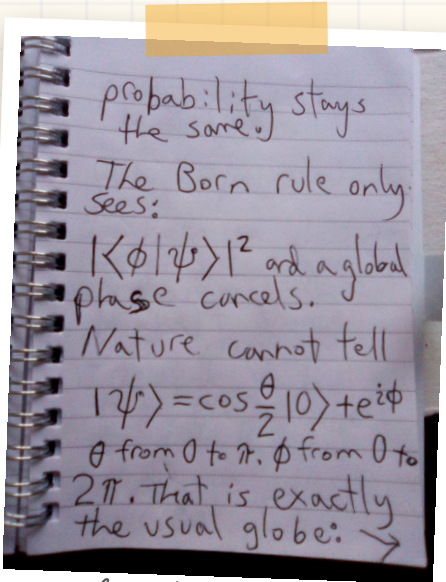
every measurement probability stays the same. The Born rule only ever sees

$$|\langle\phi|\psi\rangle|^2 \quad \text{and} \quad |e^{i\gamma}|^2 = 1,$$

so a global phase simply cancels. No experiment, ever, can tell $|\psi\rangle$ from $e^{i\gamma}|\psi\rangle$. So one more number is unphysical.

THE PUNCHLINE

Nature cannot tell.



from the notebook 📖

Three minus one leaves two.

Two continuous real numbers that describe a compact set of states: that is a **2-sphere**. Ta-da. That's why one qubit lives on a ball's surface.

$$4 \xrightarrow{-1} 3 \xrightarrow{-1} 2$$

$$\mathbb{C}^2 \rightarrow S^3 \rightarrow S^2$$

Fun aside: it has a name!



Squashing S^3 down to S^2 by forgetting the global phase is called the **Hopf fibration**. Every point on my sphere secretly has a whole little circle of phases sitting over it, all looping through one another like linked keyrings. Mathematicians think it's gorgeous. So do I.

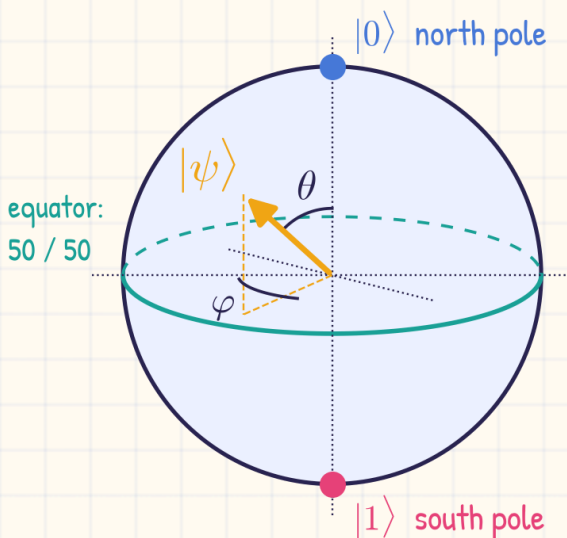
Welcome to the Globe

You can write those two leftover numbers as two angles. The standard form is:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

θ from 0 to π , φ from 0 to 2π

That is exactly the usual globe: polar angle and longitude.

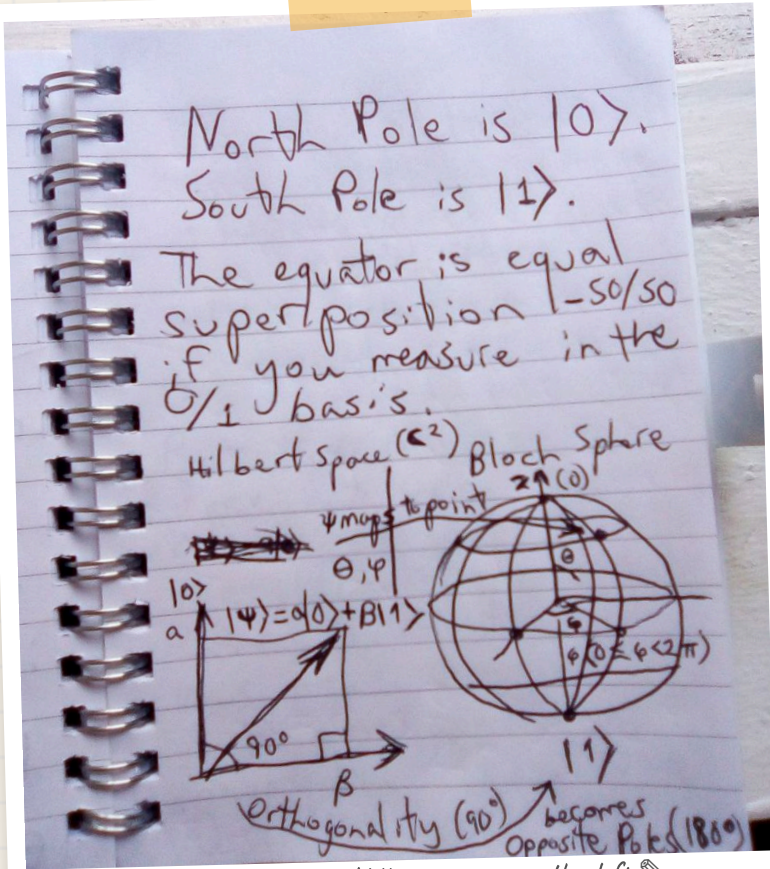


Bloch's margin note



One cosy correction: θ is the **polar angle**, measured down from the North Pole, not the latitude (which is measured up from the equator). So $\theta = 0$ is the top, $\theta = \pi/2$ is the equator, $\theta = \pi$ is the bottom. φ really is longitude, though. Full marks there.

North, South and the 50/50 Belt



her Bloch sphere, Hilbert space on the left

North Pole is $|0\rangle$. South Pole is $|1\rangle$. The equator is an equal superposition: 50/50 if you measure in the 0/1 basis.

Every point on that belt is a coin-toss; they differ only in phase, φ . And the arrow along the bottom of the sketch says it all: *orthogonality (90°) becomes opposite poles (180°)*.

Opposite Poles, Opposite Outcomes

In this picture, $|0\rangle$ is placed at the North Pole and $|1\rangle$ at the South Pole.

But this positioning does not mean they are "180° apart" in the Hilbert-space sense. Rather, the Bloch sphere uses measurement behaviour, not inner-product geometry, to place states.

$$\langle 0|1\rangle = 0 \text{ (90° as vectors)}$$



opposite poles (180° on the sphere)

A qubit in $|0\rangle$ is certain to give 0. A qubit in $|1\rangle$ is certain to give 1. Since their measurement outcomes are mutually exclusive, the Bloch sphere places them at opposite ends of the Z-axis.

Poles = states with completely opposite measurement properties.

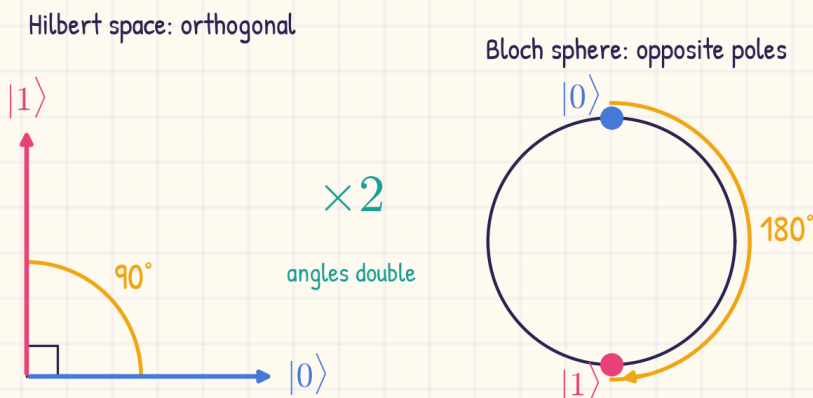


Blochy's margin note

So $|0\rangle$ and $|1\rangle$ aren't "pointing opposite ways" like two arrows in a room. They're **perfectly distinguishable**: one measurement always tells them apart, every single time. That's what "opposite" means on my sphere.

Why $\theta/2$? The Great Stretch

Why $\theta/2$ instead of θ ? Because the two basis states are orthogonal in Hilbert space (90° apart as vectors), but on the Bloch sphere we put them at opposite poles (180° apart).



Using $\theta/2$ stretches that 90° into 180° , so ordinary 3-D intuition works:

- opposite points on the sphere are opposite quantum states, and
- a 180° rotation on the sphere is a **bit flip**, $|0\rangle \leftrightarrow |1\rangle$.

That is the whole reason.



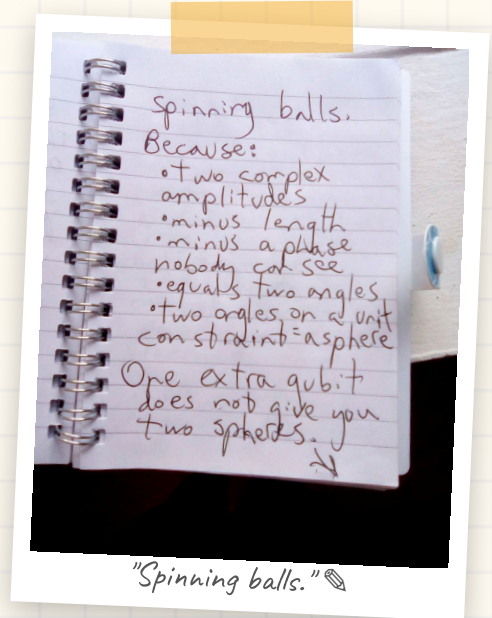
Bloch's margin note

The sphere is a map of states, not a little model of me spinning in your kitchen. Which brings us neatly to the next page...

Not Because Qubits Are Little Spinning Balls

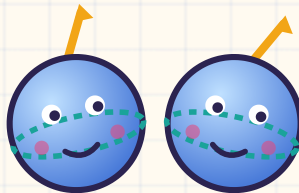
It's tempting. A sphere, an arrow, a rotation... surely the qubit is a tiny ball, twirling? No! The sphere turns up because of counting:

- *two complex amplitudes*
- *minus a length*
- *minus a phase nobody can see*
- *equals two angles*
- *two angles on a unit constraint = a sphere*



One extra qubit does not give you two spheres.

(Hold that thought. It's the cliffhanger for Chapter five.)



Two's Company: Entanglement

Two qubits live in a 4-dimensional complex space.

$$|\Psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle$$

Four complex numbers make eight real ones. Throw away the two fakes (normalisation and global phase) and you're left with **6 real parameters**. Two Bloch spheres would only give you $2 + 2 = 4$. There is no simple ball any more.

Entanglement is exactly the part that cannot be drawn as two separate points on two spheres.

The famous example is the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Neither qubit has a point of its own. Ask either one alone and you get a perfectly random answer; ask both and they always agree. A joint 4-D complex space, and states that often cannot factor into independent spheres.

2ⁿ Is the Raw Material

Two qubits can hold a superposition of four states, three of eight, n of 2^n . That exponential state space is the raw material. (Just 300 qubits means 2^{300} amplitudes: more than the number of atoms in the observable universe.)

$$n \text{ qubits} \Rightarrow 2^n \text{ amplitudes}$$



Bloch's margin note

Raw material, not free answers! When you measure n qubits you only ever get back n ordinary bits. All those amplitudes are flour, eggs and sugar; somebody still has to bake the cake.

The recipe: three ingredients

Superposition spreads the state over many possibilities, but on its own it is not enough.

Entanglement builds correlations stronger than any classical correlation.

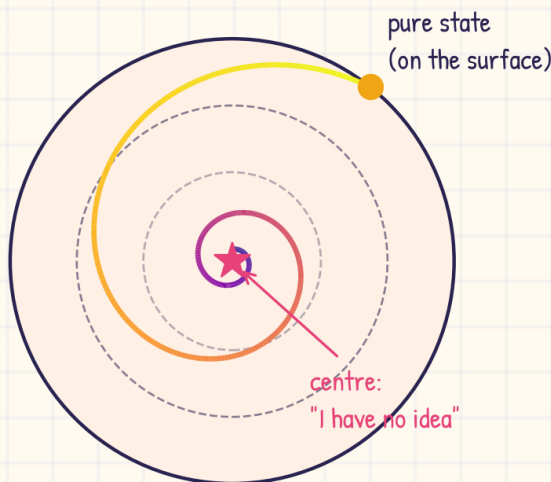
Interference: amplitudes add or cancel. Algorithms are designed so the wrong paths interfere destructively and the useful ones constructively.

Then you measure, the wavefunction collapses, and you get a classical answer.

The Ball Has an Inside

If you want the next layer, it is this: mixed states.

A qubit that has leaked information into the environment lives *inside* the ball, not only on the surface. The surface is for pure states: crisp, perfectly known. The deeper you go, the blurrier the state.



the environment keeps peeking...

The centre is "I have no idea": 50/50 and no phase left.

$$\rho = \frac{1}{2}(I + \vec{r} \cdot \vec{\sigma}), \quad |\vec{r}| = 1 \text{ pure}, \quad |\vec{r}| < 1 \text{ mixed}, \quad \vec{r} = 0 \text{ "no idea"}$$

The enemy is the environment.

Either way, the enemy is the same. The environment is always watching a little. That leaks information, which is decoherence.

Every stray photon, every wobble of heat, every tiny magnetic flutter is a little peek. Each peek pulls the point off the surface. **Decoherence is the point falling inwards.** The sphere drifts, then shrinks towards the centre (a mixed state), and the nice pure superposition dies.



That is why one qubit is easy and a useful computer is hard. One clean qubit is a solved lab project. Keeping a hundred of them coherent *and* talking to each other is the whole industry.



Blochy, feeling a bit fuzzy

Please close the fridge door. And the laser shutters. And stop looking at me. Thank you.

Look! (Oh. It's Gone.)

Measurement is the one moment the quantum world hands you something classical. You ask "0 or 1?" and the state collapses to one of the poles, with probabilities

$$P(0) = \cos^2 \frac{\theta}{2}, \quad P(1) = \sin^2 \frac{\theta}{2}$$

One shot tells you almost nothing. So you prepare, run and measure again, and again, and again. You build up the answer after many shots, because the output is probabilistic.

1 shot

"1." Thanks?

1,000 shots

≈ 70% 0, 30% 1. Oh!

Specialist engines, not replacements

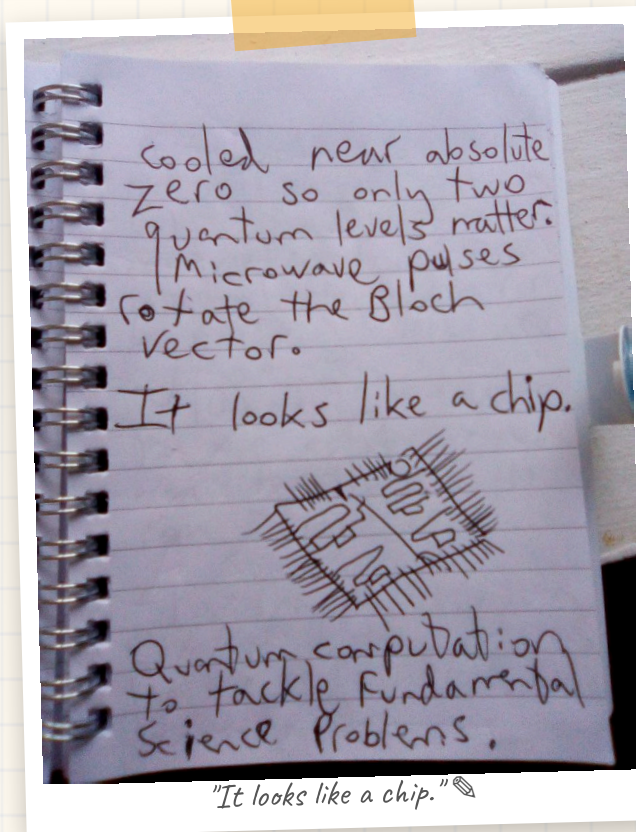
That is why quantum computers do not replace ordinary computers. They are specialist engines for a few classes of problem:

- simulating molecules and materials,
- certain optimisation and sampling tasks, and
- (eventually) breaking some public-key crypto.

They are terrible at spreadsheets, web servers and most of what you do every day.

The Hardware Zoo

Hardware today is several competing bets. First enclosure: **superconducting circuits** (IBM, Google).



Cooled near absolute zero, so only two quantum levels matter. Microwave pulses rotate the Bloch vector.

They live in a **dilution fridge**, a few thousandths of a degree above absolute zero. Fast gates, but harder to keep coherent, and lots of wiring. Her note underneath: *quantum computation to tackle fundamental science problems.*

Field Notes: Who's Who

Trapped ions (Quantinuum, IonQ): one charged atom held in an electromagnetic trap; two internal electronic (or hyperfine) levels are $|0\rangle$ and $|1\rangle$. Lasers do the rotations, and you can literally see the ion fluoresce. Slower, very high fidelity, all-to-all connectivity.

Neutral atoms (Infleqtion, QuEra, Atom Computing): easy to scale in numbers; lasers rearrange them mid-circuit. **Photonics** (PsiQuantum and others) and Microsoft's **topological** / Majorana approach are longer shots, still in the lab.

Platform	Who (her notes)	Superpower	Kryptonite
Superconducting circuits	IBM, Google	Very fast gates; built like chips	Fragile; needs a dilution fridge (~mK)
Trapped ions	Quantinuum, IonQ	Very high fidelity; all-to-all links	Slower gates
Neutral atoms	Infleqtion, QuEra, Atom Computing	Scales to big numbers; lasers rearrange atoms	Atom loss; slower cycles
Photonics	PsiQuantum and others	Light travels well; room-temp parts	Photons are hard to make interact
Topological	Microsoft (Majorana)	Errors resisted by design (in theory)	Still mostly in the lab



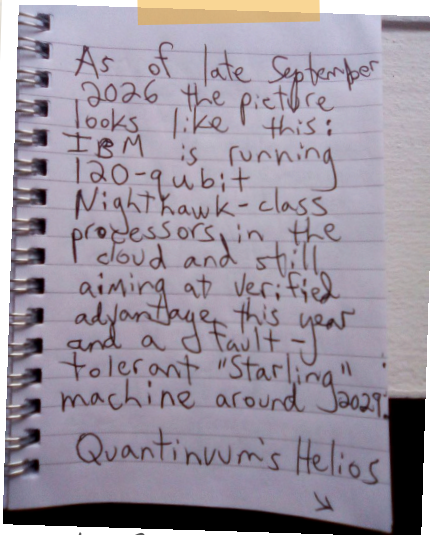
Bloch's travel tip

Same me, different houses. A qubit is any two levels you can control: an atom's energy levels, a current in a loop, a photon's polarisation. The sphere is identical; only the plumbing changes.

The Scoreboard Has Shifted

Raw physical qubit count matters less than logical qubits.

A **logical qubit** is an error-corrected group of physical qubits that behaves like one reliable qubit. In the last two years, the field finally showed that adding more physical qubits can lower the logical error rate. Google's **Willow** was the headline result. That is the threshold everyone was waiting for.



As of late September 2026 the picture looks like this: IBM is running 120-qubit Nighthawk-class processors in the cloud and still aiming at verified advantage this year and a fault-tolerant "Starling" machine around 2029.

Quantinuum's Helios

late September 2026

Google Willow

checked

105-qubit chip; showed error correction "below threshold" (bigger code, fewer logical errors), published in *Nature*, Dec 2024.

IBM

checked

Nighthawk-class processors with 120 qubits are on IBM's cloud line-up, and the fault-tolerant "Starling" (aimed at 200 logical qubits, 100 million gates) is planned for 2029, as noted in late 2026.

The 2026 Leaderboard

Her notes from late September 2026, quickly checked against the companies' own announcements in early October 2026. These are company claims and roadmaps, so read them as "what they say", not "done and dusted".

Quantinuum Helios: a trapped-ion system, often the quality leader.

checked

Helios is a 98-qubit barium-ion machine, launched commercially in November 2025.

Infleqion just showed 30 entangled logical qubits from 80 physical atoms on its Sqale neutral-atom machine (announced 24 Sept 2026).

checked

IonQ is talking Superion 256-qubit systems, with first customer deliveries planned for 2027 (announced 8 Sept 2026).

checked

US Department of Energy has a roadmap and its "Quantum Genesis" programme aimed at a scientifically useful, error-corrected machine by 2028.

checked

Germany is putting serious money into a 1,000-qubit trapped-ion push: the QUDORA-led NFQC-1k project ($\approx$ €122M, $\geq$ 1,000 physical and $\geq$ 50 logical qubits). As noted in late 2026 it had passed the outline stage and been invited to submit a full proposal, so it isn't a final award yet.

checked, with a footnote

None of this is a general-purpose quantum PC.

Useful fault-tolerant machines, with hundreds of logical qubits running millions of reliable gates, are still the late-2020s-to-2030s bet. You can't buy one to replace a GPU cluster. Claims of "quantum advantage" still have to be checked carefully against the best classical algorithms: some hold up, many are narrow or contested.

Steal Now, Decrypt Later

The other clock that is already ticking is post-quantum cryptography.

A big enough, error-corrected quantum computer could break some of today's public-key cryptography (the maths behind RSA and elliptic curves). Even if such a cryptographically relevant machine is years away, there's a sneaky problem:



record encrypted data **today** →



wait →



decrypt

it **later**

So governments and big companies such as Microsoft are migrating *now* to new, quantum-resistant algorithms, so that data stolen today can't be decrypted tomorrow.



Blochy, being reassuring

Your shopping list is safe. This is about secrets that need to stay secret for decades, and the good news is that the fix (new maths, not new physics) is already rolling out.

Scoreboard summary

- Count **logical** qubits, not just physical ones.
- Check every advantage claim against the best classical rival.
- The crypto clock is ticking regardless.

EPILOGUE

Hi There, Qubit

Charlottetown. A kitchen table, a spiral notebook, a cup of tea going cold. A familiar wheezing, groaning sound. The gull on the windowsill is not impressed.

THE DOCTOR: The scanner found a notebook. Forty-odd pages, all written by the same person, and all of them pointing at a sphere.

SHARON: Doctor Master. Hi there. Mind the tea.

THE DOCTOR: (*flipping pages*) "Four is too many. Two of them are fake." That's brilliant. That's the most honest sentence anyone's written about a qubit. Do you know how many civilisations kept all four?

SHARON: How did that go?

THE DOCTOR: They spent three centuries trying to measure a global phase. Lovely people. Very tired.

Blochy rolls out from between the pages, antenna wobbling.

BLOCHY: Is it safe? Is anyone looking?

THE DOCTOR: Only a little. The environment's always looking a little. (*he shuts the TARDIS door, firmly*) There. Pure state.

SHARON: So which is it? Are you two places at once?

BLOCHY: I'm one place. It's just a place on a sphere.

THE DOCTOR: And not because you're a little spinning ball.

BLOCHY & SHARON: Not because I'm/it's a little spinning ball!

The Doctor taps the last page. Under the leaderboard, in her handwriting: "None of this is a general-purpose quantum PC."

THE DOCTOR: She tests things before they're finished. Good. Finished things are where the trouble hides. (*to Blochy*) Coming?

BLOCHY: Only if nobody measures me on the way.

The TARDIS fades. The gull objects. On the table the notebook falls open at the North Pole, which says, quite simply: hi there.

Pocket Cheat Sheet

The qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$

The two fakes size (normalisation) and global phase $e^{i\gamma}$. Nature cannot tell.

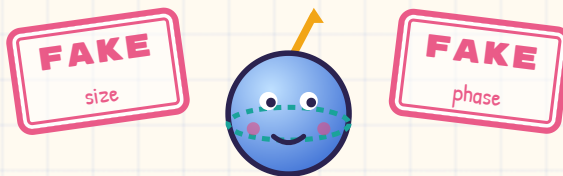
The globe $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$; θ = polar angle, φ = longitude

The stretch 90° in Hilbert space $\rightarrow 180^\circ$ on the sphere. A 180° turn is a bit flip.

The counting 1 qubit: 2 real parameters (a sphere). 2 qubits: 6 (no ball). n qubits: 2^n amplitudes of raw material, but only n bits back per measurement.

The inside Mixed states live inside the ball; the centre is "I have no idea". Decoherence is falling inwards.

The scoreboard Logical qubits beat raw counts. Check advantage claims. The PQC clock is ticking.



Made from the handwritten notebook of Sharon Epic Armando · Charlottetown, PEI ·
quantumspatial.lol

Company and roadmap notes as of late 2026; check for updates.